

previously. These are the managers who more often trade at and into the close than in the morning hours. This is done in order to transact as close to the closing price as possible and to minimize tracking error compared to the closing price, since the fund will be valued based on the day's closing price. Fourth, there has been a dramatic increase in exchange traded funds (ETFs). ETFs are used for many different reasons. For example, ETFs are used to gain certain market exposures or to hedge very short-term risk. And very often investors who were hedging short-term market risk net out those positions at the end of the day, resulting in increased trading volume. ETF trading has also caused an increase in trading volume towards the end of the day due to the creation and redemption of these ETFs. While we have not found evidence suggesting correlation between overall ETF and stock volumes, we have found a statistically significant relationship between the shift in intraday volume trading and ETF volume. Thus, ETF trading has played a part in shifting the stock's intraday volume profile towards the close and away from the open. Fifth, the increase in closing imbalance data and the investor's ability to offset end of day imbalances (as well as easy and direct access to this information) has helped improve the price discovery process. This coupled with less end of day price volatility allows funds an easier time to achieve market prices at the close than previously and hence leads to more trading at the close. Finally, over the last few years there has been a decrease in quantitative investment strategies. Quant managers have reduced leverage of their portfolios and the general economic climate has not provided these managers with as much profiting opportunity as in years past. Since quant managers do not currently have a strong need to achieve particular market prices they are able to utilize a more passive execution style such as VWAP rather than utilizing a front-loading execution style such as IS or arrival price.

To the extent that there is another shift away from Index and ETF trading back to active management and quantitative styles, we are likely to see a corresponding shift in intraday volume profiles with more volume trading at the open and less dramatic spiking towards the close. Portfolio manager investment styles have a dramatic effect on when volumes occur throughout the day (Figure 2.5).

Volatility

- Measured as the average high-low percentage price range in each 15 minute trading period.
- Does not currently follow its historical U-shaped profile. Intraday volatility is higher at the open than mid-day and only increases slightly into the close.